

Follicular lymphoma

The prototype indolent lymphoma — **incurable when advanced, but usually a disease measured in many years**. Treat symptoms and burden (GELF), not the scan; **POD24 and transformation** are the events that change prognosis. Old grade 3B is now **follicular large B-cell lymphoma — treat as DLBCL**

Compiled from the Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin — follicular lymphoma · NCCN · primary trial reports verified against PubMed, September 2026.

BIOLOGY AND CLASSIFICATION

t(14;18) IGH::BCL2 ~85% of classic FL	BCL2 juxtaposed to the IGH enhancer → BCL2 overexpression blocks apoptosis . Germinal-center phenotype: CD20+, CD10+, BCL6+, BCL2+; CD5–, CD23– (usually). CREBBP / KMT2D / EZH2 mutations are common secondary hits
Classic FL (cFL) <i>WHO-HAEM5</i>	Old grades 1, 2 and 3A. Grading is now optional — poorly reproducible and of little value for individual treatment. Grade 3A is managed as FL
Follicular large B-cell lymphoma	WHO-HAEM5 name for old grade 3B (sheets of centroblasts, no centrocytes; often CD10– and t(14;18)–). Treat as DLBCL with R-CHOP. ICC 2022 keeps the name “FL grade 3B”
Variants that should not be over-treated	Pediatric-type FL (young, localized, BCL2–, excellent outcome) · duodenal-type FL (low dissemination risk) · in situ follicular B-cell neoplasm (incidental)

WORKUP

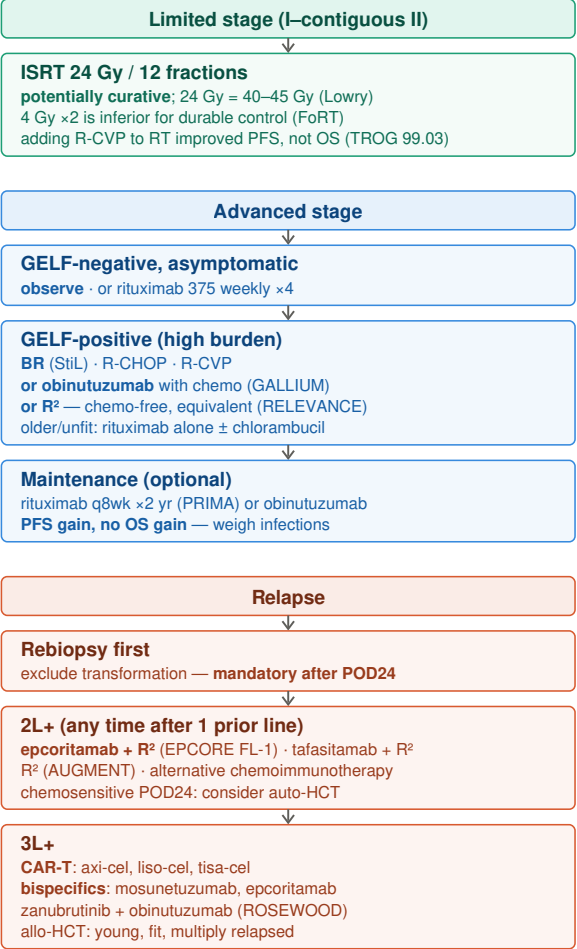
- **Excisional or generous core biopsy** — architecture and centroblast content cannot be judged on FNA
- **PET/CT** for staging; Lugano / Ann Arbor. Marrow biopsy when it would change management (e.g. apparent limited stage before RT)
- CBC, CMP, **LDH**, **β2-microglobulin**, **HBsAg + anti-HBc** (reactivation with anti-CD20), HCV, HIV
- **Biopsy the PET-hottest lesion** if SUV is discordant or one node grows out of step — it is the site most likely to have transformed

FLIPI — “NOLASH”, 1 POINT EACH

Risk	%	5-yr OS	10-yr OS
Low (0–1)	36	91%	71%
Intermediate (2)	37	78%	51%
High (3–5)	27	53%	36%

Nodal areas >4 · LDH >ULN · Age ≥60 · Stage III–IV · Hgb <12 (Solal-Céligny, Blood 2004; pre-rituximab era). **FLIPI-2**: age >60, β2M >ULN, marrow involved, Hgb <12, largest node >6 cm — 5-yr PFS 80% / 51% / **19%**. **Neither index tells you when to treat** — GELF does.

TREATMENT PATHWAY



GELF — ANY ONE = HIGH BURDEN, TREAT

- ≥3 nodal sites, each ≥3 cm
- Any nodal or extranodal mass ≥7 cm
- B symptoms or other disease-related symptoms
- Splenomegaly · effusion or ascites · organ compression
- **Cytopenias** (ANC <1,000 or platelets <100,000) · leukemic phase >5,000/μL

Brice, JCO 1997. **Meeting none = observe**. Watch-and-wait has never lost survival against early treatment; rituximab ×4 delays the need for next therapy (3-yr no new treatment **88% vs 46%**, HR 0.21; Ardeshtna, Lancet Oncol 2014) but has not improved OS.

FIRST-LINE EVIDENCE

Question	Answer
BR vs R-CHOP <i>StiL NHL1</i>	BR: median PFS 69.5 vs 31.2 mo , HR 0.58, far less toxicity (no alopecia, less neuropathy). No OS difference
Obinutuzumab vs rituximab <i>GALLIUM</i>	7-yr PFS 63.4% vs 55.7% ; OS identical (88.5% vs 87.2%) . More serious AEs (48.9% vs 43.4%) and infusion reactions. With bendamustine: more grade 3–5 infections, second cancers and fatal events — prolonged T-cell depletion
R² vs R-chemo <i>RELEVANCE</i>	Negative superiority trial — 6-yr PFS 60% vs 59%, OS 89% both. An equivalent chemo-free option: less neutropenia, more rash
Maintenance <i>PRIMA</i>	Median PFS 10.5 vs 4.1 yr at 9 yr follow-up; no OS benefit ; grade 2–4 infections 39% vs 24%
Low burden: retreat or maintain? <i>RESORT</i>	Rituximab retreatment at progression = maintenance for time to treatment failure (3.9 vs 4.3 yr), using 4 vs 18 doses

POD24 is the prognostic event that matters. ~20% progress within 2 years of first-line chemoimmunotherapy; their **5-yr OS is ~50% vs ~90%** (FLIPI-adjusted HR 6.4; Casulo, JCO 2015). Implications: **rebiopsy** (a large share are transformed), avoid simply repeating the same chemo, and move early to **R²-based triplets, CAR-T or bispecifics**; consolidate chemosensitive relapse with auto-HCT in selected patients.

PEARLS

- **FLIPI prognosticates; GELF decides**. A high FLIPI with no GELF criterion is still a reason to observe
- **No first-line upgrade has improved survival** — not obinutuzumab, not maintenance, not R². Choose on toxicity and patient priorities
- **Old grade 3B (FLBL) gets R-CHOP**, not BR; grade 3A is treated as FL
- **Rebiopsy every progression that looks different** — POD24, a discordant node, a hot PET lesion, a rising LDH
- **Think ahead to CAR-T** before giving bendamustine to a high-risk patient

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RELAPSED / REFRACTORY — THE AGENTS

Agent · US status	Evidence
R² (lenalidomide + rituximab) <i>Regular approval (FL, MZL; 2019)</i>	AUGMENT (phase 3, FL + MZL, not rituximab-refractory): PFS 39.4 vs 14.1 mo , HR 0.46; 5-yr update OS HR 0.59 . Benefit carried by FL — the MZL subgroup was not significant
Epcoritamab + R² <i>Approved 2L+, Nov 2025</i>	EPCORE FL-1 (phase 3, ≥1 prior line): ORR 95% vs 79% , PFS HR 0.21 , 16-mo PFS 85.5% vs 40.2%. Grade ≥3 AEs 90% vs 68%; CRS low grade (gr 2 in 5%). OS immature
Tafasitamab + R² <i>Approved 2L+, Jun 2025</i>	inMIND (phase 3, double-blind): median PFS 22.4 vs 13.9 mo , HR 0.43. No CRS — a fit for patients unsuited to T-cell engagers
Mosunetuzumab (CD20×CD3, IV) <i>Accelerated, 3L+</i>	Single-arm phase 2, 3L+: CR 60% ; fixed duration — 8 cycles if CR, up to 17 . CRS 44%, mostly grade 1–2. CELESTIMO (+ lenalidomide vs R ² , 2L+) met PFS — topline only, Sept 2026
Epcoritamab (CD20×CD3, SC) <i>Regular approval, 3L+ (Nov 2025)</i>	EPCORE NHL-1 FL cohort, 3L+: ORR 82% , CR 62.5% ; continuous until progression. 3-step-up dosing cut CRS to 49% (none ≥grade 3)
Axi-cel <i>Accelerated, 3L+</i>	ZUMA-5 (single arm, 3L+): ORR 94% FL; median PFS 62.2 mo (all iNHL) at 5-yr follow-up; few relapses after 30 mo. Grade ≥3 neurotoxicity 19%
Liso-cel <i>Accelerated, 3L+ (May 2024)</i>	TRANSCEND FL : 3L+ ORR 97% , CR 94%; 2L high-risk (POD24) ORR 96%. CRS 58% (grade ≥3 1%)
Tisa-cel <i>Accelerated, 3L+</i>	ELARA : CR 69%, ORR 86%; no grade ≥3 CRS
Zanubrutinib + obinutuzumab <i>Accelerated, 3L+ (Mar 2024)</i>	ROSEWOOD (randomized phase 2 vs obinutuzumab): ORR 69% vs 46% , median PFS 28.0 vs 10.4 mo. Confirmatory MAHOGANY (vs R ²) ongoing

No head-to-head data between CAR-T and bispecifics. CAR-T gives the most durable single-shot responses; bispecifics are off-the-shelf and outpatient. Recent bendamustine blunts CAR-T expansion and outcome (ZUMA-5) — plan the sequence.

WITHDRAWN — DO NOT REACH FOR THESE

Agent	What happened
Tazemetostat (EZH2)	Accelerated approval 2020 (3L+ EZH2-mutant, or no alternative). Withdrawn from the US market 2026 : in SYMPHONY-1 (+ R ²), hematologic second cancers (MDS/AML, B-ALL) in 5.7% vs 0% , 3 deaths
PI3K inhibitors	Idelalisib, duvelisib FL indications withdrawn 2021–22; umbralisib withdrawn 2022 (excess deaths); copanlisib withdrawn Nov 2023 (CHRONOS-4 negative). Class problem: infections, colitis, pneumonitis, hepatitis, and an OS signal
Ibrutinib	Never approved for FL; confirmatory SELENE (FL/MZL) missed PFS

TRANSFORMATION

- **10-yr cumulative risk ~5–8%** in the rituximab era (5.2% with rituximab vs 8.7% without; Aristotle, Lancet Haematol 2018)
- **Suspect it** with discordant nodal growth, new B symptoms, rising LDH, hypercalcemia, new extranodal disease, or **a PET lesion much hotter than the rest**
- **Biopsy the PET-hottest site** — a random node can miss it. Check **MYC**: with the existing BCL2 rearrangement a MYC break makes it **HGBL with MYC + BCL2**
- **Treat as DLBCL**: anthracycline-naïve → R-CHOP; already had R-CHOP → salvage + auto-HCT or **CAR-T** (axi-cel, liso-cel and tisa-cel labels include DLBCL arising from FL)

HOW TO CHOOSE AT RELAPSE

- **Late relapse, low burden**: rituximab again, or R². **Symptomatic, first relapse**: epcoritamab + R² is now the strongest randomized 2L data; tafasitamab + R² if a T-cell engager is not suitable
- **POD24 or double-refractory** (anti-CD20 + alkylator): favor **CAR-T** (liso-cel has 2L POD24 data) or a bispecific; auto-HCT for chemosensitive early relapse
- **Frail**: rituximab alone, R², or zanubrutinib + obinutuzumab. **Palliation of a symptomatic node**: 2 × 2 Gy

REGIMEN DOSING

Regimen	Dosing
Rituximab	375 mg/m² weekly ×4 ; maintenance 375 q8 wk ×2 yr (PRIMA). SC rituximab 1400 mg after a first full IV dose
BR	Bendamustine 90 mg/m² d1+2 + rituximab 375 d1, q28d ×6
R-CHOP / R-CVP	Rituximab 375, cyclophosphamide 750, vincristine 1.4 (max 2 mg) d1, prednisone 40 mg/m ² d1–5, ± doxorubicin 50 — q21d ×6 (CHOP) or ×8 (CVP)
Obinutuzumab	1000 mg flat, C1 d1, 8, 15 , then d1 of each cycle (C2–6 with benda, C2–8 with CHOP/CVP); maintenance 1000 mg q2 mo ×2 yr
R²	Lenalidomide 20 mg PO d1–21 q28d ×12 (10 mg if CrCl 30–59); rituximab 375 weekly C1 then d1 C2–5
Epcoritamab + R²	Epcoritamab 48 mg SC after step-up, weekly C1–3, then q4wk C4–12 (fixed duration) + R ² as above
Epcoritamab mono	SC 0.16 → 0.8 → 3 → 48 mg (C1 d1, 8, 15, 22), 48 mg weekly C2–3, q2wk C4–9, then q4wk until progression
Mosunetuzumab	IV 1 → 2 → 60 mg (C1 d1, 8, 15), 60 mg C2 d1, then 30 mg q21d; stop after 8 cycles if CR , else up to 17
Tafasitamab + R²	Tafasitamab 12 mg/kg d1, 8, 15, 22 C1–3, then d1+15 C4–12 (q28d) + R ²
Zanubrutinib + obinutuzumab	Zanubrutinib 160 mg BID continuous + obinutuzumab 1000 mg (C1 d1, 8, 15; C2–6 d1; then q8wk)
ISRT	24 Gy / 12 fx curative intent · 2 × 2 Gy palliative

mg/m² unless stated. Premedicate every anti-CD20 infusion (acetaminophen, antihistamine, steroid for the first).

TOXICITY AND MONITORING

- **HBV**: screen HBsAg + anti-HBc before any anti-CD20; **entecavir prophylaxis** if either positive, continued for ≥12 months after the last dose
- **Bendamustine → profound, prolonged CD4 lymphopenia** (months to years). Give **PJP prophylaxis (TMP-SMX) and VZV prophylaxis (acyclovir/valacyclovir)**; watch for CMV. Worse when followed by maintenance anti-CD20 — think twice about stacking them. Also hampers later CAR-T manufacturing
- **Obinutuzumab**: infusion reactions ~60% (mostly C1 d1) — premedicate, slow first infusion, hold antihypertensives. More neutropenia and infection than rituximab
- **Bispecifics**: CRS clusters around the **first full dose** of step-up. Corticosteroid premedication, hydration, close monitoring after the first full dose; tocilizumab for grade ≥2. ICANS uncommon. Continuous B-cell depletion → **hypogammaglobulinemia and serious infections**: IVIG if recurrent, PJP + antiviral prophylaxis, vaccinate beforehand
- **R²: VTE** — aspirin 81 mg if no other risk, LMWH/DOAC if high risk; **rash** (often grade 1–2; antihistamine, hold/reduce), neutropenia (50% grade 3–4 in AUGMENT), tumor flare, diarrhea. Dose by renal function
- **CAR-T**: CRS and ICANS (REMS removed June 2025: stay near the center and no driving for **2 weeks**), prolonged cytopenias, hypogammaglobulinemia; T-cell malignancy boxed warning
- **Zanubrutinib**: bleeding (hold 3–7 days peri-procedure), AF, hypertension, neutropenia

KEY TRIALS

SiIL NHL1	BR > R-CHOP for PFS; no OS difference
GALLIUM	Obinutuzumab: PFS gain, no OS gain
RELEVANCE	R ² = R-chemo (negative superiority)
PRIMA	Maintenance: PFS doubled, no OS
ForT	24 Gy beats 4 Gy for local control
AUGMENT	R ² vs R at relapse: PFS HR 0.46
EPCORE	Epcoritamab + R ² : PFS HR 0.21
FL-1	
inMIND	Tafasitamab + R ² : PFS HR 0.43